PRACTICAL – 11

Write a program for Floyd-Warshal algorithm

Aim:
To write a C++ program to implement the Floyd–Warshall algorithm for finding the shortest paths between all pairs of vertices in a weighted graph.

Objective:
To calculatetheshortest distance between every pair of vertices using the dynamic programmingtechnique.

Algorithm:

1. Start.
2. Initializethe distance matrix using the given graph's edge weights.
3. Set thedistance from each vertex to itself as 0.
4. Considereach vertex k as an intermediate vertex.
5. For everypair of vertices i and j, update the distance using:
   [dist[i][j] = \min(dist[i][j], dist[i][k] + dist[k][j])]
6. Repeatuntil all vertices have been considered as intermediate vertices.
7. Displaythe shortest distance matrix.
8. Stop.

Code:

```
V = 4
INF = 99999

graph = [
   [0, 5, INF, 10],
   [INF, 0, 3, INF],
   [INF, INF, 0, 1],
   [INF, INF, INF, 0]
]

dist = [row[:] for row in graph]

for k in range(V):
   for i in range(V):
      for j in range(V):
```

```
            if (dist[i][k] != INF and
                dist[k][j] != INF and
                dist[i][k] + dist[k][j] < dist[i][j]):

                dist[i][j] = dist[i][k] + dist[k][j]

    print("Shortest Distance Matrix:")

    for i in range(V):
        for j in range(V):
            if dist[i][j] == INF:
                print("INF", end="\t")
            else:
                print(dist[i][j], end="\t")
        print()

    print("EN.No: 92460118673")
```

Output:

```
PS C:\Users\vatsalya\Downloads> & 'C:\Users\vatsalya\AppData\Local\Programs\Python\Python315\python.exe' 'c:\Users\vatsalya\
Downloads\main(2).py'
Shortest Distance Matrix:
0       5       8       9
INF     0       3       4
INF     INF     0       1
INF     INF     INF     0
EN.No: 92460118673
PS C:\Users\vatsalya\Downloads>
```

Time Complexity:

Space Comp	*Time Complexity*	*Operation*
O(V^2)	*O(V^2)*	**Initialize distance matrix**
O(1)auxiliary	*O(V^3)*	**Floyd–Warshall algorithm**
O(V^2)	*O(V^3)*	**Overall**